

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Advanced Math

17

10 answers · Rationale-rich · Teach from doc

Q1**D**

D. The estimated initial population for this species and area in 2008

Choice D is correct. Substituting 0 for x in the given equation yields $f(0) = 3,000(0.75)^0$, which is equivalent to $f(0) = 3,000(1)$, or $f(0) = 3,000$. It's given that the function estimates the species' population x years after 2008, so it follows that the estimated population of the species is 3,000 in 2008. Therefore, the best interpretation of 3,000 in this context is the estimated initial population for this species and area in 2008.

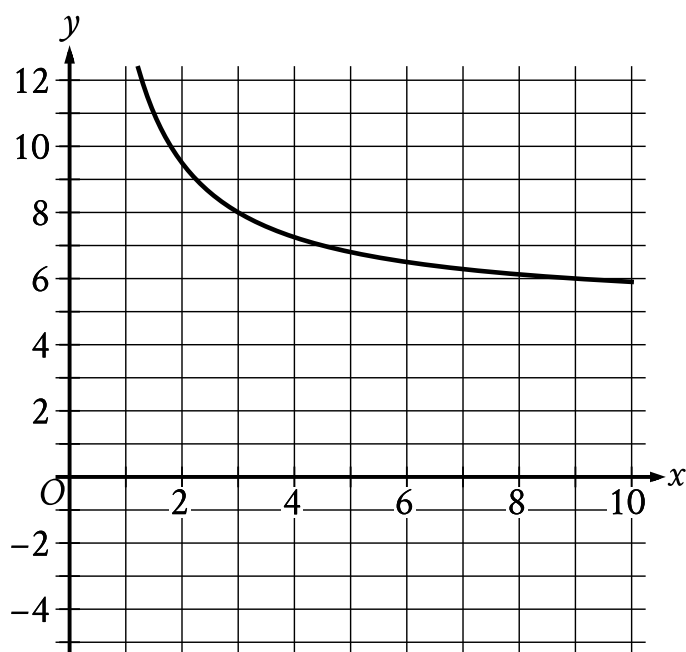
Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. The estimated percent decrease in the population for this species and area each year after 2008 is 25%, not 3,000.

Choice C is incorrect. The estimated population for this species and area 8 years after 2008 is $3,000(0.75)^8$, or approximately 300, not 3,000.

Q2

D



D.

- Moving from left to right:
 - The curve is shown in quadrant 1.
 - The curve trends down sharply to x equals 3.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 5.
- The curve passes through the following points:
 - $(2, \frac{19}{2})$
 - $(3, 8)$
 - $(6, \frac{13}{2})$

Choice D is correct. It's given that the graph of the rational function f is shown, where $y = fx$ and $x \geq 0$. The graph shown passes through the point 3, 3. It follows that when the value of x is 3, the value of fx is 3. When the value of fx is 3, the value of $fx + 5$ is $3 + 5$, or 8. Therefore, the graph of $y = fx + 5$ passes through the point 3, 8. Of the given choices, choice D is the only graph that passes through the point 3, 8 and is therefore the graph of $y = fx + 5$.

Choice A is incorrect. This is the graph of $y = fx - 5$, rather than $y = fx + 5$.

Choice B is incorrect. This is the graph of $y = \frac{fx}{5}$, rather than $y = fx + 5$.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3

C

C. $p(t) = 24(1.162)^t$

Choice C is correct. This exponential growth model can be written in the form $p(t) = A(1 + r)^t$, where $p(t)$ is the population t years after 1788, A is the initial population, and r is the yearly growth rate, expressed as a decimal. Since there were 24 jackrabbits in Australia in 1788, $A = 24$. Since the number of jackrabbits increased by an average of 16.2% each year, $r = 0.162$. Therefore, the equation that best models this situation is $p(t) = 24(1.162)^t$.

Choices A, B, and D are incorrect and may result from misinterpreting the form of an exponential growth model.

Q4

3600

The correct answer is 3,600. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting 0 for x in the given equation yields $y = 3,600(a)^0$. Since a is positive, this equation is equivalent to $y = 3,600(1)$. Therefore, when $x = 0$, the value of y is 3,600. Thus, the predicted number of bacteria initially measured in the growth medium is 3,600.

Q5**B****B. 14.50**

Choice B is correct. It's given that the function h estimates that the object is 3,364 feet above the ground when it's dropped at $t = 0$. Substituting 3,364 for ht and 0 for t in the function h yields $3,364 = -16t^2 + b$, or $3,364 = b$. Substituting 3,364 for b in the function h yields $ht = -16t^2 + 3,364$. When the object hits the ground, its height will be 0 feet above the ground. Substituting 0 for ht in $ht = -16t^2 + 3,364$ yields $0 = -16t^2 + 3,364$. Adding $16t^2$ to each side of this equation yields $16t^2 = 3,364$. Dividing each side of this equation by 16 yields $t^2 = 210.25$. Since the object will hit the ground at a positive number of seconds after it's dropped, the value of t can be found by taking the positive square root of each side of this equation, which yields $t = 14.50$. It follows that the function estimates the object will hit the ground approximately 14.50 seconds after being dropped.

Choice A is incorrect. The function estimates that 7.25 seconds after being dropped, the object's height will be $-16(7.25)^2 + 3,364$ feet, or 2,523 feet, above the ground.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**D**

D. $f(x) = 3,000(0.98)^x$

Choice D is correct. Because the change in the number of students decreases by the same percentage each year, the relationship between the number of students and the number of years can be modeled with a decreasing exponential function in the form $f(x) = a(1 - r)^x$, where $f(x)$ is the number of students, a is the number of students in 2015, r is the rate of decrease each year, and x is the number of years since 2015. It's given that 3,000 students were enrolled in 2015 and that the rate of decrease is predicted to be 2%, or 0.02. Substituting these values into the decreasing exponential function yields $f(x) = 3,000(1 - 0.02)^x$, which is equivalent to $f(x) = 3,000(0.98)^x$.

Choices A, B, and C are incorrect and may result from conceptual errors when translating the given information into a decreasing exponential function.

Q7**A**

A. 0, 11

Choice A is correct. The x -coordinate of any y -intercept of a graph is 0. Substituting 0 for x in the given equation yields $g(0) = 11 \cdot \frac{1}{12}^0$. Since any nonzero number raised to the 0th power is 1, this gives $g(0) = 11 \cdot 1$, or $g(0) = 11$. The y -intercept of the graph is, therefore, the point 0, 11.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**C**

C. $fx = 2331.70^{\frac{x}{2}}$

Choice C is correct. It's given that on April 1, there were 233 views of the advertisement. It's also given that every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. This situation can be represented by an exponential function of the form $fx = a1 + \frac{r}{100}^{\frac{x}{c}}$, where a is the number of views on April 1 and every c days after April 1, the number of views had increased by $r\%$ of the number of views c days earlier. It follows that $a = 233$, $r = 70$, and $c = 2$. Substituting 233 for a , 70 for r , and 2 for c in the equation $fx = a1 + \frac{r}{100}^{\frac{x}{c}}$ yields $fx = 2331 + \frac{70}{100}^{\frac{x}{2}}$, or $fx = 2331.70^{\frac{x}{2}}$.

Choice A is incorrect. This function gives the predicted number of views for an advertisement for which every 2 days, the number of views was 70%, rather than increased by 70%, of the number of views 2 days earlier.

Choice B is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, the number of views was 70% of the number of views $\frac{1}{2}$ days earlier, rather than an advertisement for which every 2 days, the number of views had increased by 70% of the number of views 2 days earlier.

Choice D is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, rather than every 2 days, the number of views had increased by 70% of the number of views $\frac{1}{2}$ days earlier, rather than 2 days earlier.

Q9**B**

B. $y = -\frac{1}{10}xx - 4x + 5^2$

Choice B is correct. Each of the given choices is an equation of the form $y = -\frac{1}{10}xx - a^mx + b^n$, where a , b , m , and n are positive constants. In the xy -plane, the graph of an equation of this form has x -intercepts at $x = 0$, $x = a$, and $x = -b$. The graph shown has x -intercepts at $x = 0$, $x = 4$, and $x = -5$. Therefore, $a = 4$ and $b = 5$. Of the given choices, only choices A and B have $a = 4$ and $b = 5$. For an equation in the form $y = -\frac{1}{10}xx - a^mx + b^n$, if all values of x that are less than $-b$ or greater than a correspond to negative y -values, then the sum of all the exponents of the factors on the right-hand side of the equation is even. In the graph shown, all values of x less than -5 or greater than 4 correspond to negative y -values. Therefore, the sum of all the exponents of the factors on the right-hand side of the equation $y = -\frac{1}{10}xx - 4^mx + 5^n$ must be even. For choice A, the sum of these exponents is $1 + 1 + 1$, or 3 , which is odd. For choice B, the sum of these exponents is $1 + 1 + 2$, or 4 , which is even. Therefore, $y = -\frac{1}{10}xx - 4x + 5^2$ could be the equation of the graph shown.

Choice A is incorrect. For the graph of this equation, all values of x less than -5 correspond to positive, not negative, y -values.

Choice C is incorrect. The graph of this equation has x -intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x -intercepts at $x = -5$, $x = 0$, and $x = 4$.

Choice D is incorrect. The graph of this equation has x -intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x -intercepts at $x = -5$, $x = 0$, and $x = 4$.

Q10

C

C. $px = 2,0001.03^x$

Choice C is correct. It's given that a function p estimates that there were 2,000 animals in a population in 1998 and that each year from 1998 to 2010, the number of animals in this population increased by 3% of the number of animals in the population the previous year. It follows that this situation can be represented by the function $px = a1 + \frac{r}{100}^x$, where px is the estimated number of animals in the population x years after 1998, a is the estimated number of animals in the population in 1998, and each year the estimated number of animals increased by $r\%$. Substituting 2,000 for a and 3 for r in this function yields $px = 2,0001 + \frac{3}{100}^x$, or $px = 2,0001.03^x$.

Choice A is incorrect. This function represents a population in which each year the number of animals increased by 200%, not 3%, of the number of animals in the population the previous year.

Choice B is incorrect. This function represents a population in which each year the number of animals increased by 97%, not 3%, of the number of animals in the population the previous year.

Choice D is incorrect. This function represents a population in which each year the number of animals decreased, rather than increased, by 3% of the number of animals in the population the previous year.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank